

Water Quality Sensor Noise Classification based on Drone Movement using Machine Learning

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Abstract— Water pollution caused by human activities and environmental changes has become a critical issue. The challenges of directly analyzing water quality in remote or inaccessible aquatic environments highlight the importance of combining drones and water quality sensors. However, external factors, such as wind-induced drone movements, can introduce noise into sensor readings, compromising the accuracy of data collection. This study focuses on addressing the patterns of drone movements and their impact on sensor noise to enhance the reliability of water quality monitoring. In this research, experimental results demonstrate that machine learning models effectively analyzed the movement patterns, with detailed performance metrics for each algorithm. XGBoost (extreme gradient boosting), Random Forest, Decision Tree, and Logistic Regression were among the methods that were assessed. The Random Forest model outperformed the others, including XGBoost, Decision Tree, and Logistic Regression, with an accuracy of 97%, compared to 96% and 85%, respectively. The findings identify Random Forest as the most effective algorithm for analyzing movement patterns and mitigating noise in sensor data, thereby improving the overall reliability of water quality assessments.

Keywords—Drone movement, water quality sensor, machine learning, and noise sensor

I. INTRODUCTION

Coastal habitats may be negatively impacted by human activities involving ports, rivers, the construction of ports and piers, land reclamation, beach sand filling, and offshore energy producing fields [1][2]. A reduction in water quality, harm to marine life, excessive algal photosynthesis, and sediment erosion or buildup that changes the shape of the coast are a few examples of these effects [3][4].

Additional risks may be introduced by post-construction activities such as shipping, loading, unloading, and permanent anchoring, as well as the management of industrial and urban waste disposal [5]. Water quality can also be greatly impacted by human-caused phenomena like global warming and climate change [6]. Additional variations in coastal water quality can also result from physical factors such as freshwater inflow, land drainage, precipitation, wind, wave action, and the presence of plant and animal life in coastal zones [7][8]. In light of these possible dangers, environmental monitoring especially of water quality is crucial to preserving a healthy and sustainable ecosystem under a variety of stressors. One of the main forces behind the creation and application of suitable monitoring equipment is the preservation of the environment and human health.

Current water quality monitoring has evolved with various methods, ranging from periodic field testing to cloud-based

monitoring systems [9][10]. However, promising technologies like sensor-equipped drones have not been optimally utilized. The use of sensor drones in water monitoring has numerous advantages, especially in reaching difficult-to-access areas and significantly enhancing the efficiency and accuracy of data collection [11][12]. Therefore, implementing sensor drones in water quality monitoring systems is crucial for obtaining more comprehensive and precise data.

One significant advancement in the efficient and precise collecting of environmental data is drone-based water quality monitoring [13]. Drones with sensors are able to assess important aspects of water quality, including temperature, dissolved oxygen, pH, and pollutant concentrations. Nevertheless, there are still a lot of obstacles to overcome, especially with regard to sensor stability when in flight [14]. Excessive sensor movement due to drone instability is one of the main challenges in data collection [15]. When drones face external disturbances such as strong winds, sudden changes in altitude, or sharp maneuvers, the sensors may experience drastic swings. This movement can result in inaccurate data or even render it scientifically invalid, leading to errors in water quality analysis [16].

Mechanical dampening or other hardware components are frequently used in traditional methods to drone stability problems. The flexibility, accuracy, and dependability of these approaches are constrained, particularly in dynamic environmental settings [17]. Machine learning technology provides a creative and effective solution in this situation. Predictive models that can categorize and analyze drone motions and sensor readings in real time are made possible by machine learning algorithms [18]. This method improves the precision of data collecting while fortifying the system's general validity and dependability.

The novelty of this paper is the classification of drone movement and the analysis of how they affect the functioning of water quality sensors. Drone flight syslog data can be carefully examined to find trends in sensor dynamics by applying machine learning algorithms. This analysis makes it easier to categorize sensor data disruptions like noise, especially those brought on by outside influences like wind. In addition to offering diagnostic insights into the reasons behind these disruptions, particularly those pertaining to sensor or drone movement, the model can successfully discriminate between noisy and clean data. Pilots and system specialists can quickly resolve problems with drone stability, environmental effects, or sensor data quality thanks to the quick detection of noise. The precision, stability, and dependability of data gathering systems are greatly increased

by this machine learning-driven analysis, providing a strong basis for more effective and accurate drone applications in a range of environmental circumstances.

II. LITERATUR REVIEW

A. Excessive Drone Movement

Data collection in areas that are challenging to reach by hand is made easier by the use of drones in water quality monitoring [19]. However, erratic or excessive motion while in flight may result in inaccurate sensor readings. Unstable drone control or environmental conditions like wind might cause excessive drone motions, which can cause external interference in analog sensors. The drone's orientation is changed by these extreme motions, which also raise the possibility of variations that don't match the real aquatic environmental conditions [20][21].

Oscillations and sudden position changes in drones cause unstable sensor measurements, producing values that deviate from actual conditions or even failing to acquire data. This situation is challenging because unstable movements create noise in the data and increase the need for stabilization or filtering techniques in the system [22]. With these data fluctuations due to excessive movement, precision in water quality measurements becomes difficult to achieve. Therefore, controlling excessive drone movement becomes crucial to ensure that collected data accurately reflects environmental conditions.

B. Sensor Anomalies during Excessive Movement

Excessive movement triggers anomalies in analog sensors, such as pH, turbidity, TDS, and water temperature sensors. Analog sensors are vulnerable to interference due to position changes or excessive vibration, generating anomalous signals that affect data quality. When these sensors experience unstable movement, the resulting data is often distorted, and the emerging noise makes analysis less valid. These data anomalies require mitigation strategies to ensure data accuracy under dynamic flight conditions [23].

Sensor readings are heavily influenced by sudden movements as these sensors are sensitive to orientation changes [24][25]. To address this, several approaches such as stabilization platforms and real-time signal filtering algorithms can be implemented. Understanding sensor anomaly characteristics in the context of excessive movement helps identify the need for measurement system improvements and data correction methods, ensuring final results better represent actual environmental conditions.

C. Machine Learning Classification

Particularly in environmental monitoring applications, the use of machine learning techniques to categorize anomalies in drone sensor data is becoming increasingly popular. Finding unusual patterns in drone sensor data has been successfully accomplished using machine learning techniques [26]. Machine learning makes it easier to automatically detect distorted data and distinguish it from legitimate data by training models using historical datasets that contain anomalous patterns brought on by erratic movements. Crucial machine learning methods including feature selection, exploratory data analysis (EDA), and hyperparameter tuning are used to further improve this process [27][28][29][30]. By reducing data disturbance brought on by drone movements, these techniques enhance

the precision and dependability of water quality monitoring systems.

This study looks at four classification methods and uses performance metrics to assess each one's effectiveness. The model's ability to detect all real positive instances is evaluated by recall (or sensitivity); accuracy measures the proportion of accurate predictions among all instances; precision measures the proportion of true positive predictions among all positive predictions; and the F1-score provides a harmonic mean of precision and recall, balancing the trade-off between the two. The study ensures a full comparison of the categorization methods to determine their benefits and drawbacks in various scenarios by employing these evaluation criteria.

This study focuses on describing logistic regression in [31]. Logistic regression is a machine learning technique that does not require data to satisfy statistical assumptions, such as the independence of observations and the absence of multicollinearity among independent variables. It can therefore be applied to a wide variety of data types. A machine learning technique called logistic regression doesn't need data to meet statistical presumptions such the independence of observations and the lack of multicollinearity between independent variables. Because of this, it may be used with many kinds of data. The probability $P(y=1|X)$ in logistic regression is the chance that, given the independent variables X , the dependent variable y will formula 1. A linear combination of input variables (z) can be converted into a probability using logistic regression and the sigmoid function. Here, z is calculated as a combination of the intercept (β_0), regression coefficients ($\beta_1, \beta_2, \dots, \beta_n$) and independent variables ($x_1, x_2, \dots, x_n$). The sigmoid function ensures that the output probability lies within the range of 0 and 1, making it easily interpretable for data-driven decision-making. Formula (1) show calculation for logistic regression.

$$P(y = 1|X) = \frac{1}{1 + e^{-z}} \quad (1)$$

In [32], the random forest and decision tree algorithms are discussed. Random forest is a machine learning technique that combines predictions from several decision trees to boost accuracy and decrease overfitting. A random subset of data and features is used to construct each tree, and majority voting or average is used to determine the final forecast. In a random Forest, the final predicted class y for an input x is determined by mode ($\{f_i(x)\}$), which represents the most frequently occurring class among the predictions from all the trees in the forest. Here, $\{f_i(x)\}$ is the set of predicted classes from each individual tree i , where $i = 1, 2, \dots, k$ and k is the total number of trees in the forest. This majority voting mechanism ensures robust and accurate predictions by aggregating the outputs of multiple decision trees. Formula (2) show main calculation for random forest algorithm.

$$y = \text{mode}\{f_i(x)\} \quad (2)$$

Decision tree is a simple machine learning model that bases its judgments on a tree-like structure. A regression value or class label is represented by each leaf node, a feature test by each internal node, and a test result by each branch. In this calculation, entropy—a measure of disorder or unpredictability at a node—is used to evaluate the purity of a

split in a decision tree. The percentage of instances at the node that belong to class i is calculated using p_i , where n is the total number of classes. Formula (3) show main calculation for decision tree algorithm.

$$Entropy = -\sum_{i=1}^n p_i \log_2(p_i) \quad (3)$$

In [33], Extreme Gradient Boosting, or XGBoost, is a high-performance machine learning technique that creates trees one after the other to fix mistakes in earlier trees. This paper discusses XGBoost. In XGBoost, the loss function $l(y_i, y_i')$ measures the difference between the predicted value y_i and the actual value y_i , ensuring the model minimizes prediction errors. Additionally, the regularization term $\Omega(f_k)$ penalizes the complexity of the model f_k , helping to prevent overfitting and improve generalization by balancing model performance and simplicity. Formula (4) show main calculation for XGBoost algorithm.

$$Obj = \sum_{i=1}^n l(y_i, y_i') + \sum_{k=1}^K \Omega(f_k) \quad (4)$$

III. PROPOSED METHOD

This study focuses on the classification system used to analyze the effects of drone movements for water quality sensor readings. Fig. 1. illustrates the proposed method utilized in this research.

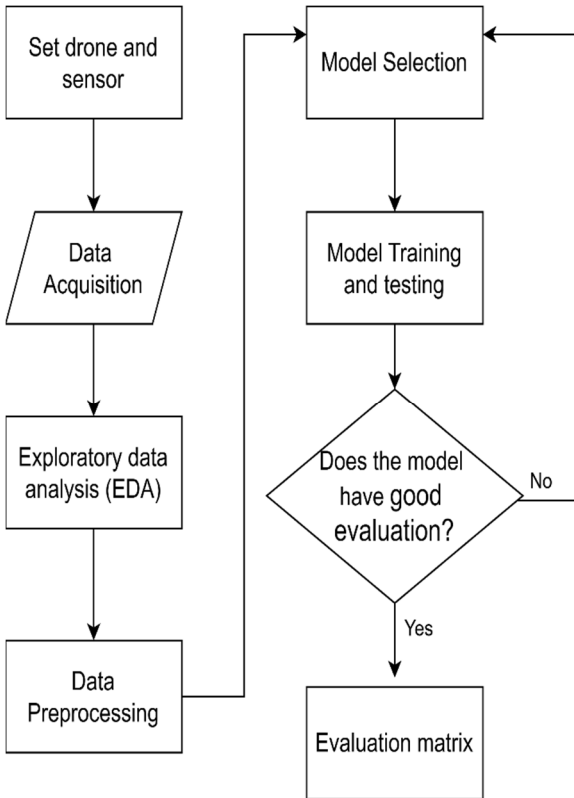


Fig. 1. Proposed Method

A. Drone and Sensor Setting

To ensure effective operation and control, the drone system is made up of a number of interconnected parts. Fig. 2. show drone system communication from this research by combining inputs from the GPS, winch, and power modules with a current sensor, the flight controller functions as the primary processor. Four rotors power the propulsion system. A LiPo (lithium polymer) battery provides power, which is controlled by a LiPo charger control system to ensure safe charging. Two telemetry modules enable communication: Telemetry 1 pairs the drone with the Ground Control Station, enabling real-time data sharing with a computer for planning and monitoring, and Telemetry 2 pairs with the Pilot Control for manual navigation. The GPS module provides precise positioning for autonomous operations, and the winch module is used for payload handling, such as deploying or retrieving sensors during tasks. Together, this setup enables both autonomous and manual operations with reliable power management and communication.

The sensor system is designed to record and track water quality parameters using an integrated setup. The sensor system used in this study is shown in Fig. 3. The TDS (Total Dissolved Solids) sensor detects the concentrations of dissolved particles, the temperature sensor records the temperature of the water, the turbidity sensor measures the clarity of the water, and the pH sensor determines the acidity or alkalinity of the water. Its four primary sensors are these. These sensors are linked to a microcontroller, which acts as the central processing unit, to collect data from the sensors and ensure proper system operation. Additional components include an accelerometer for movement or vibration detection, a real-time clock module for accurate data time-stamping, and a battery to power the entire device. At the data storage module, data is stored for further analysis.

B. Data Acquisition

A drone and sensors are used to immediately collect data as part of the data acquisition process. The sensors are installed on the drone in this configuration, which is hung and subject to outside factors, particularly wind. Analyzing the variations in sensor data between the drone's steady, stationary position and when it is exposed to wind is the goal of the study. This method makes it possible to examine how the behavior of the drone and the sensor readings under various circumstances relate to one another. Eight hours are allotted for data collection, with four dedicated to steady readings and the other four influenced by wind conditions. The goal of this research is to clearly understand how external elements, like wind, affect drone stability and sensor performance.

C. Data Sheet Information

The data sheet shows the outcomes that were directly acquired through data collection. Four different classes are used to categorize the collected data: Class 0 indicates that there was no wind and no noise was detected; Class 1 indicates that there was wind but noise was detected; Class 2 indicates that there was wind but no noise was detected; and Class 3 indicates that there was wind and noise. Sensor calibration is used to determine the classification; to account for changes and guarantee proper categorization, the stability

value of the sensor is changed by deducting the tolerance value.

D. Model Selection

A study that can identify complex connections between variables is required since every stable sensor reading exhibits unique values and patterns. Machine learning is therefore essential to this research. This study employs four primary methods to classify data according on specific patterns: XGBoost with hyperparameter tweaking from random searchCV, Random Forest, Decision Tree, and Logistic Regression. To ensure accurate, sensitive, and balanced predictions, each method's performance is evaluated using metrics like as accuracy, precision, recall, and F1-Score. Finding the optimal algorithm for analyzing sensor data with various patterns is the aim of this approach.

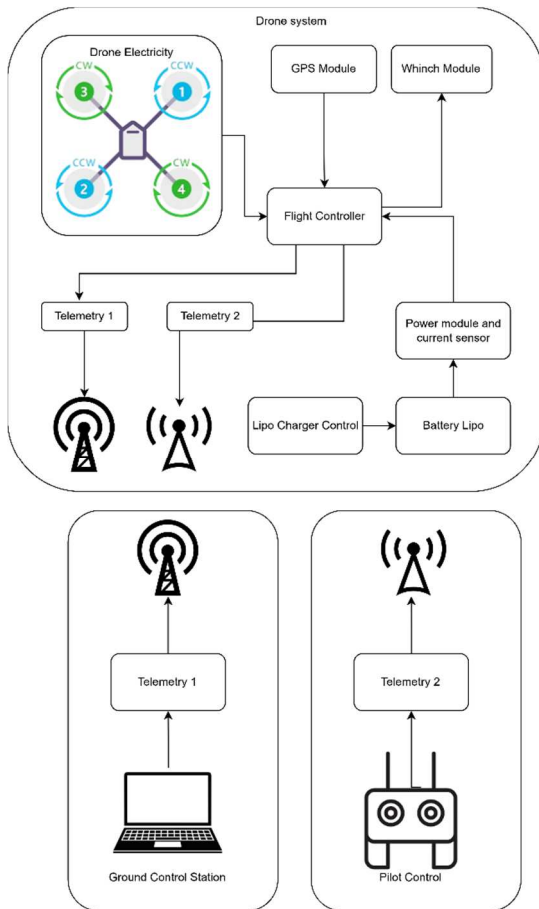


Fig. 2. Drone system communication

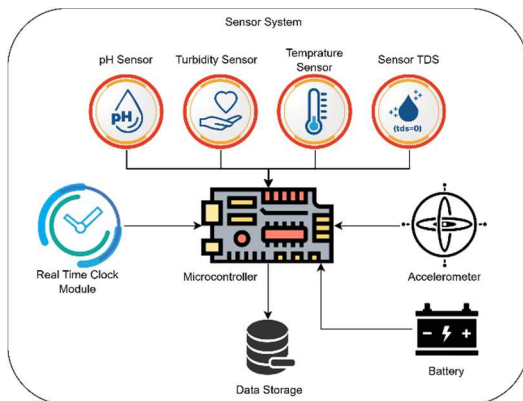


Fig. 3. Sensor system communication

IV. RESULT AND DISCUSSION

A. Data Result Information

Drones and sensors provide the data for analysis, and timestamps are used to synchronize the results. Four classes have been assigned to the data: No wind and no noise are indicated by Class 0; no wind and noise are indicated by Class 1; no wind but noise is detected by Class 2; and both wind and noise are identified by Class 3. The labels are distributed as follows: There are 13,494 data points total, including 7,205 recordings under Label 0 (no wind and no noise), 74 records under Label 1 (no wind but noise), 780 records under Label 2 (wind but no noise), and 5,435 records under Label 3 (wind plus noise). Fig. 4. illustrates the sensor readings, highlighting the conditions when the sensor detects noise and when it operates without detecting any noise.

An example of data gathered from the drone, including accelerometer readings, pitch, roll, and yaw, is shown in Table 1. In the meantime, accelerometer readings and sensor data from four different types of sensors temperature, TDS, pH, and turbidity. A thorough understanding of environmental conditions based on observed physical and chemical characteristics is provided by the combination of drone and sensor data. Establishing a thorough connection between wind, noise, and other sensory variables is the goal of this comprehensive investigation.

B. Classification Result

Based assessment metrics, Table 2 show the evaluation matric, the classification results show how well four models Random Forest, XGBoost, Logistic Regression, and Decision Tree perform when 10-fold cross-validation and hyperparameter tuning via Random Search CV are used. With 97% and 96% accuracy on the test data, respectively, Random Forest and XGBoost demonstrated exceptional performance. Precision 0.99, recall 0.97, and F1-scores 0.98 additionally, random forest was the top-performing model, demonstrating its remarkable capacity to reduce classification errors. With only a few small misclassifications such as 59 cases of class 3 being mistakenly classified as class 1 the random forest confusion matrix shows extremely accurate predictions for every class, including 2082 right predictions for class 0 and 1107 for class 3. XGBoost performed similarly with an F1-score of 0.97, but its confusion matrix revealed slightly more errors, such as 106 instances of class 3 misclassified as class 1, though it remains a robust model overall.

On the other hand, when compared to ensemble-based models, Logistic Regression and Decision Tree performed worse. With just 750 cases accurately predicted and 271 incorrectly classified as class 1, logistic regression struggled with large misclassifications despite achieving respectable accuracy (85%). Decision Tree demonstrated a greater number of misclassifications for specific classes, such as 84 cases of class 3 being mistakenly forecasted as class 1, even though it achieved an accuracy comparable to XGBoost (96%). Random Forest showed greater generalization with its ideal hyperparameter combination, such as `n_estimators: 150` and `max_depth: 30`, as shown by its outstanding evaluation metrics and balanced distribution of right predictions in the confusion matrix. These results establish Random Forest as the best method for classifying this dataset.

TABLE I. DATA SAMPLE FORM DRONE AND SENSOR SYSTEM

timestamp	Drone Data						Sensor Data							Label
	roll (°)	pitch (°)	yaw (°)	xacc (m/s ²)	yacc (m/s ²)	zacc (m/s ²)	pH	TDS (ppm)	Temp (°C)	Turbidity (NTU)	AccelX (m/s ²)	AccelY (m/s ²)	AccelZ (m/s ²)	
2024-11-14 05:36:06	-0.01	-0.10	-0.81	1.02	0.11	9.67	6.56	560.53	28.81	10.00	0.49	0.41	9.86	0
2024-11-21 01:48:41	0.02	-0.21	-3.01	2.35	0.18	9.45	7.97	583.40	27.87	10.00	0.48	0.59	9.84	0
2024-11-20 22:02:40	0.02	-0.21	-3.05	2.34	0.18	9.40	8.31	589.03	27.87	10.00	0.46	0.69	9.83	1
2024-11-20 22:38:17	0.02	-0.21	-3.08	2.32	0.17	9.42	8.30	591.87	28.06	10.00	0.48	0.58	9.85	1
2024-11-14 03:17:38	0.02	-0.10	-0.85	1.01	-0.02	9.71	6.50	564.42	27.87	10.00	0.44	0.42	9.75	2
2024-11-14 03:26:24	0.02	-0.10	-0.88	1.01	0.08	9.72	6.50	564.42	27.94	10.00	0.48	0.40	9.84	2
2024-11-20 21:11:51	0.03	-0.21	2.24	2.45	0.24	9.36	8.06	593.90	29.06	10.00	0.45	0.89	9.82	3
2024-11-20 22:17:03	0.01	-0.19	-2.87	2.28	-0.13	9.40	8.29	591.87	28.00	10.00	0.43	0.59	9.88	3

TABLE II. EVALUATION MATRIX RESULT

Model	Best Parameter	CV-Accuracy	Test Accuracy	Precision	Recall	F1 Score
Random Forest	{n_estimators: 150, min_samples_split: 2, min_samples_leaf: 1, 'classifier_max_depth': 30, 'bootstrap': False}	0.97	0.97	0.99	0.97	0.98
XGBoost	{subsample: 0.7, n_estimators: 50, 'max_depth': 3, 'learning_rate': 0.3, 'gamma': 0.1, colsample_bytree: 0.8}	0.97	0.96	0.99	0.96	0.97
Logistic Regression	{solver: 'liblinear', 'penalty': 'l2', 'C': 10}	0.85	0.85	0.92	0.85	0.87
Decision Tree	{min_samples_split: 10, min_samples_leaf: 2, max_depth: None}	0.96	0.96	0.98	0.96	0.97

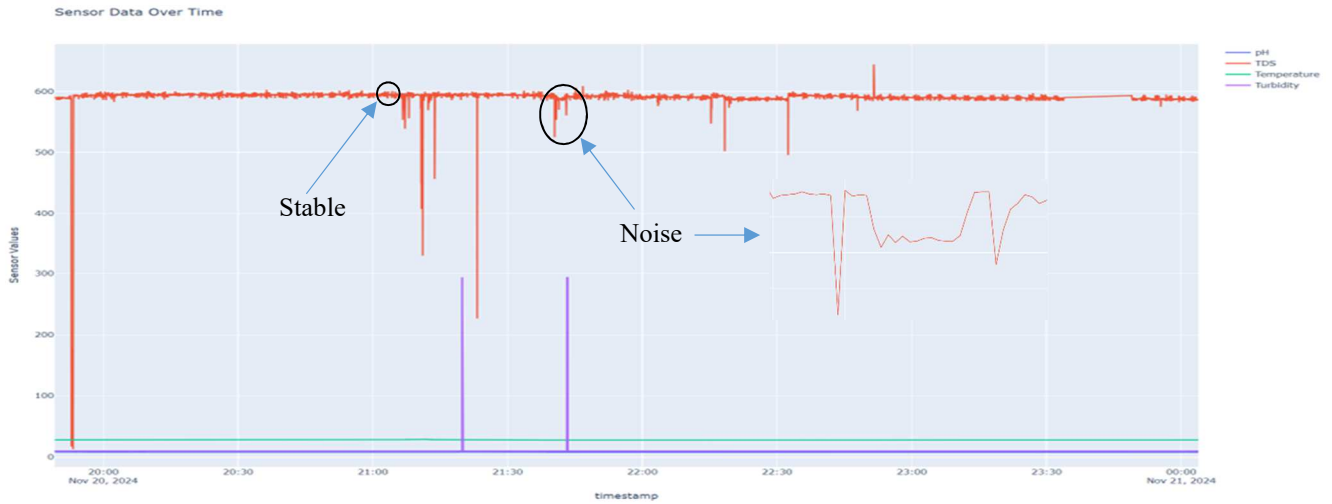


Fig. 4. Sensor Data

V. CONCLUSION

Drones with water quality sensors are a great way to keep an eye on the state of the water. Faster and more thorough observations are possible because to this integration. According to the experimental study, the Random Forest (RF) model performed exceptionally well in identifying and distinguishing real data from noise, achieving the greatest accuracy of 97%. With a 96% accuracy rate, the XGBoost model came in second, followed by the Decision Tree model, which also scored 96% but had somewhat worse precision and recall than XGBoost. The lower accuracy of 85% for logistic regression, on the other hand, demonstrated its limits when dealing with noisy data. Because of its exceptional accuracy and resilient noise management, Random Forest is the most dependable and efficient model for this use case.

Future study can concentrate on additional simulations to address internal difficulties connected to drone operations. This method seeks to more thoroughly examine the internal and external variables influencing sensor readings so that researchers can more successfully choose reliable data.

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REFERENCES

- [1] K. Orviku, H. Tonisson, R. Aps, J. Kotta, I. Kotta, G. Martin, U. Suursaar, R. Tamsalu, and V. Zalesny, "Environmental impact of port construction: Port of Sillamäe case study (Gulf of Finland, Baltic Sea)," in 2008 IEEE/OES US/EU-Baltic International Symposium, 2008, pp. 1-10, doi: 10.1109/BALTIC.2008.4625538.
- [2] K. Slavik, C. Lemmen, W. Zhang, O. Kerimoglu, K. Klingbeil, and K. W. Wirtz, "The large-scale impact of offshore wind farm structures on pelagic primary productivity in the southern North Sea," *Hydrobiologia*, vol. 845, no. 1, pp. 35-53, 2019, doi: 10.1007/s10750-018-3653-5.
- [3] T. Pang, X. Wang, R. A. Nawaz, G. Keefe, and T. Adekanmbi, "Coastal erosion and climate change: A review on coastal-change process and

- modeling," *Ambio*, vol. 52, no. 12, pp. 2034–2052, 2023, doi: 10.1007/s13280-023-01901-9.
- [4] C. J. Gobler, "Climate Change and Harmful Algal Blooms: Insights and perspective," *Harmful Algae*, vol. 91, no. December 2019, p. 101731, 2020, doi: 10.1016/j.hal.2019.101731.
- [5] A. Tayeb, M. R. Chellali, A. Hamou, and S. Debbah, "Impact of urban and industrial effluents on the coastal marine environment in Oran, Algeria," *Marine Pollutan Bulletin*, vol. 98, no. 1–2, pp. 281–288, 2015, doi: 10.1016/j.marpolbul.2015.07.013.
- [6] J. Xiong, Y. Zheng, J. Zhang, F. Quan, H. Lu, and H. Zeng, "Impact of climate change on coastal water quality and its interaction with pollution prevention efforts," *Journal of Environment Management*, vol. 325, no. PB, p. 116557, 2023, doi: 10.1016/j.jenvman.2022.116557.
- [7] S. Bhattarai, P. Parajuli, and A. Linhoss, "Integrated Modeling Approach to Assess Freshwater Inflow Impact on Coastal Water Quality," *Water (Switzerland)*, vol. 16, no. 21, pp. 1–18, 2024, doi: 10.3390/w16213012.
- [8] G. Griggs and B. G. Reguero, "Coastal Adaptation to Climate Change and Sea-Level Rise. Water 2021, 13, 2151. <https://doi.org/10.3390/w13162151> Academic," 2021.
- [9] M. M. Grand, G. S. Clinton-Bailey, A. D. Beaton, A. M. Schaap, T. H. Johengen, M. N. Tamburri, D. P. Connelly, M. C. Mowlem, and E. P. Achterberg, "A lab-on-chip phosphate analyzer for long-term In Situ monitoring at fixed observatories: Optimization and performance evaluation in estuarine and oligotrophic coastal waters," *Frontiers in Marine Science*, vol. 4, no. AUG, pp. 1–16, 2017, doi: 10.3389/fmars.2017.00255.
- [10] F. Adamo, F. Attivissimo, C. G. C. Carducci, and A. M. L. Lanzolla, "A smart sensor network for sea water quality monitoring," *IEEE Sensor Journal*, vol. 15, no. 5, pp. 2514–2522, 2015, doi: 10.1109/JSEN.2014.2360816.
- [11] L. De Keukelaere, R. Moelans, E. Knaeps, S. Sterckx, I. Reusen, D. De Munck, S. G. H. Simis, A. M. Constantinescu, A. Scricciu, G. Katsouras, W. Mertens, P. D. Hunter, E. Spyarakos, and A. Tyler, "Airborne Drones for Water Quality Mapping in Inland, Transitional and Coastal Waters—MapEO Water Data Processing and Validation," *Remote Sensing*, vol. 15, no. 5, 2023, doi: 10.3390/rs15051345.
- [12] P. D. Dai, D. M. Quang, N. M. H. Giang, and L. H. Hieu, "Efficient Water Quality Monitoring Using Unmanned Aerial Vehicles and Internet of Thing Technologies," In *2024 16th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)*, pp. 1–6, 2024, doi: 10.1109/ECAI61503.2024.10606972.
- [13] R. Zhang, Z. Wang, X. Li, Z. She, and B. Wang, "Water Quality Sampling and Multi-Parameter Monitoring System Based on Multi-Rotor UAV Implementation," *Water (Switzerland)*, vol. 15, no. 11, 2023, doi: 10.3390/w15112129.
- [14] K. R. I. Sanim, M. Kalaitzakis, B. Kosaraju, Z. Kitzhaber, C. English, N. Vitzilaos, M. Myrick, M. Hodgson, and T. Richardson, "Development of an Aerial Drone System for Water Analysis and Sampling," *2022 International Conference on Unmanned Aircraft Systems (ICUAS)*, pp. 1601–1607, 2022, doi: 10.1109/ICUAS54217.2022.9836122.
- [15] J. Szczepaniak, B. Szlachetko, and M. Lower, "The Influence of Temporal Disturbances in EKF Calculations on the Achieved Parameters of Flight Control and Stabilization of UAVs," *Sensors*, vol. 24, no. 12, 2024, doi: 10.3390/s24123826.
- [16] V. Sadhu, K. Anjum, and D. Pompili, "On-Board Deep-Learning-Based Unmanned Aerial Vehicle Fault Cause Detection and Classification via FPGAs," *IEEE Transaction on Robotics*, vol. 39, no. 4, pp. 3319–3331, 2023, doi: 10.1109/TRO.2023.3269380.
- [17] J. Huang, M. Li, Y. Zhang, L. Mu, Z. Ao, and H. Gong, "Fault Detection and Classification for Sensor Faults of UAV by Deep Learning and Time-Frequency Analysis," In *2021 40th Chinese Control Conference (CCC)*, vol. 2021-July, pp. 4420–4424, 2021, doi: 10.23919/CCC52363.2021.9550141.
- [18] S. Silalahi, T. Ahmad, H. Studiawan, E. Anthi, and L. Williams, "Severity-Oriented Multiclass Drone Flight Logs Anomaly Detection," *IEEE Access*, vol. 12, no. March, pp. 64252–64266, 2024, doi: 10.1109/ACCESS.2024.3396926.
- [19] H. Yang, J. Kong, H. Hu, Y. Du, M. Gao, and F. Chen, "A Review of Remote Sensing for Water Quality Retrieval: Progress and Challenges," *Remote Sensing*, vol. 14, no. 8, 2022, doi: 10.3390/rs14081770.
- [20] B. Bayat, N. Crasta, A. Crespi, A. M. Pascoal, and A. Ijspeert, "Environmental monitoring using autonomous vehicles: a survey of recent searching techniques," *Current opinion in biotechnology*, vol. 45, no. 645141, pp. 76–84, 2017, doi: 10.1016/j.copbio.2017.01.009.
- [21] J. H. Ryu, "UAS-based real-time water quality monitoring, sampling, and visualization platform (UASWQP)," *HardwareX*, vol. 11, p. e00277, 2022, doi: 10.1016/j.ohx.2022.e00277.
- [22] J. Xu, P. Shi, C. C. Lim, C. Cai, and Y. Zou, "Reliable Tracking Control for Under-Actuated Quadrotors with Wind Disturbances," *IEEE Transactions on systems, man, and cybernetics: systems*, vol. 49, no. 10, pp. 2059–2070, 2019, doi: 10.1109/TSMC.2017.2782662.
- [23] F. Esmaili, E. Cassie, H. P. T. Nguyen, N. O. V. Plank, C. P. Unsworth, and A. Wang, "Anomaly Detection for Sensor Signals Utilizing Deep Learning Autoencoder-Based Neural Networks," *Bioengineering*, vol. 10, no. 4, pp. 1–30, 2023, doi: 10.3390/bioengineering10040405.
- [24] A. Lanata, E. P. Scilingo, E. Nardini, G. Loriga, R. Paradiso, and D. De-Rossi, "Comparative evaluation of susceptibility to motion artifact in different wearable systems for monitoring respiratory rate," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 378–386, 2010, doi: 10.1109/TITB.2009.2037614.
- [25] S. Kang, A. Paul, and G. Jeon, "Reduction of mixed noise from wearable sensors in human-motion estimation," *Computers & Electrical Engineering*, vol. 61, pp. 287–296, 2017, doi: 10.1016/j.compeleceng.2017.05.030.
- [26] B. Wang, Z. Wang, L. Liu, D. Liu, and X. Peng, "Data-Driven Anomaly Detection for UAV Sensor Data Based on Deep Learning Prediction Model," In *2019 Prognostics and System Health Management Conference (PHM-Paris)*, pp. 286–290, 2019, doi: 10.1109/PHM-Paris.2019.00055.
- [27] J. Galvan, A. Raja, Y. Li, and J. Yuan, "Sensor Data-Driven UAV Anomaly Detection using Deep Learning Approach," In *MILCOM 2021-2021 IEEE Military Communications Conference (MILCOM)*, vol. 2021-November, pp. 589–594, 2021, doi: 10.1109/MILCOM52596.2021.9653036.
- [28] D. Sunaryono, R. Sarno, J. Siswanto, A. B. Raharjo, S. I. Sabilla, R. I. Susilo, and K. Rekha, "Enhanced Salp Swarm Algorithm Based on Convolutional Neural Network Optimization for Automatic Epilepsy Detection," *Journal of Theoretical and Applied Information Technology*, vol. 100, no. 19, pp. 5615–5623, 2022.
- [29] D. R. Wijaya, R. Sarno, and E. Zulaika, "Sensor array optimization for mobile electronic nose: Wavelet transform and filter based feature selection approach," *International Review on Computers and Software*, vol. 11, no. 8, pp. 659–671, 2016, doi: 10.15866/irecos.v11i8.9425.
- [30] F. A. Haq, R. Sarno, R. Abdullah, T. C. Amri, A. F. Septiyanto, and K. R. Sungkono, "Feature Selection of Photoplethysmograph Data in Machine Learning," *2023 International Conference on Artificial Intelligence in Information and Communication (ICAIIIC)*, pp. 315–320, 2023, doi: 10.1109/ICAIIIC57133.2023.10067116.
- [31] S. Nusinovic, Y. C. Tham, M. Y. C. Yan, D. S. W. Ting, J. Li, C. Sabanayagam, T. Y. Wong, and C.-Y. Cheng, "Logistic regression was as good as machine learning for predicting major chronic diseases," *Journal of Clinical Epidemiology*, vol. 122, pp. 56–69, 2020, doi: 10.1016/j.jclinepi.2020.03.002.
- [32] Z. Sun, G. Wang, P. Li, H. Wang, M. Zhang, and X. Liang, "An improved random forest based on the classification accuracy and correlation measurement of decision trees," *Expert Systems with Applications*, vol. 237, no. PB, p. 121549, 2024, doi: 10.1016/j.eswa.2023.121549.
- [33] A. Ogunleye and Q. G. Wang, "XGBoost Model for Chronic Kidney Disease Diagnosis," *IEEE/ACM transactions on computational biology and bioinformatics*, vol. 17, no. 6, pp. 2131–2140, 2020, doi: 10.1109/TCBB.2019.2911071.